TUGAS ALGORITMA PEMROGRAMAN

PENJUMLAHAN MATRIS ORDO 2X2

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1. Deskrpsi Masalah

Diberikan dua matriks A dan Bberordo 2×2. Tentukan hasil penjumlahan matriks $C = A + B$. Rumus:

$$C_{ij} = A_{ij} + B_{ij}$$

2. Identifikasi Input – Proses – Output

- **Input:** Matriks Adan Matriks B(masing-masing berordo 2×2)
- **Proses:** Menjumlahkan elemen-elemen yang bersesuaian dari kedua matriks
- **Output:** Matriks hasil penjumlahan C

3. Pseudocode

```
BEGIN

    INPUT matriks A (2x2)

    INPUT matriks B (2x2)

    FOR i = 1 TO 2

        FOR j = 1 TO 2

            C[i][j] ← A[i][j] + B[i][j]

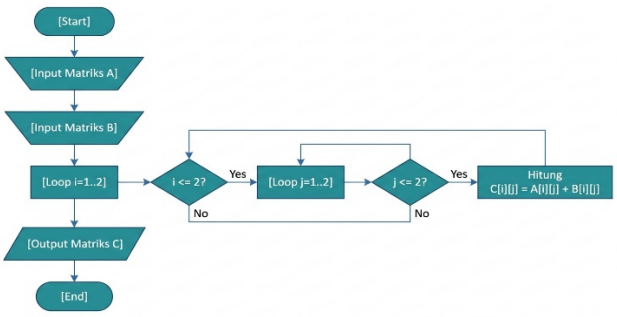
        END FOR

    END FOR

    OUTPUT matriks C

END
```

4. Flowchart



5. **Test Case**

   - **Test Case 1:**

     A = [[1, 2], [3, 4]]

     B = [[5, 6], [7, 8]]

     Hasil: C = [[6, 8], [10, 12]]

   - **Test Case 2:**

     A = [[2, 0], [1, -3]]

     B = [[-2, 4], [3, 5]]

     Hasil: C = [[0, 4], [4, 2]]

6. **Implementasi Python**

```
# Program penjumlahan matriks 2x2

# Algoritma dan Pemrograman | S1 Pendidikan Matematika FKIP
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def tambah_matriks(A, B):

    C = [[0, 0], [0, 0]]

    for i in range(2):

        for j in range(2):

            C[i][j] = A[i][j] + B[i][j]

    return C

# Input dari pengguna

print("Masukkan elemen matriks A (2x2):")

A = [[int(input(f"A[{i}][{j}]: ")) for j in range(2)] for i in
range(2)]

print("Masukkan elemen matriks B (2x2):")

B = [[int(input(f"B[{i}][{j}]: ")) for j in range(2)] for i in
range(2)]
```

```
# Proses dan Output

C = tambah_matriks(A, B)

print("Hasil penjumlahan matriks C adalah:")

for row in C:

    print(row)

# Test case tambahan

print("Test Case 1:", tambah_matriks([[1,2],[3,4]],
[[5,6],[7,8]]))  # [[6,8],[10,12]]

print("Test Case 2:", tambah_matriks([[2,0],[1,-3]], [[-
2,4],[3,5]])) # [[0,4],[4,2]]
```